

Question

Pain and motor severity
co-occur in PD

Which comes first?
Or neither?

Dopaminergic dose
changes with both

Method

RI-CLPM
separates between-person
from within-person

Mixed model + IPCW
level vs slope,
attrition weighted

MSM (IPTW) + g-formula
time-varying confounding
by prior exposure

Finding

Stable trait covariation
 $r = 0.175$
No temporal precedence

Level effect, not slope
+1.00 points/pain point
Slope $p = 0.19$

Attenuates by ~37%
but persists
1.71 (0.72 to 2.67)